

GOVERNMENT INTERVENTIONS IN MARKETS PROBLEM SET—SOLUTIONS

ECON101—INTRODUCTION TO MICROECONOMICS.

SPRING 2026

PROF. ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

1 Taxes and Subsidies

1.1 Taxes

(a) $p^e = \frac{\alpha - \gamma}{\beta + \delta}$

(b) $Q^e = \frac{\delta\alpha + \beta\gamma}{\beta + \delta}$

(c) $p^d = \frac{\alpha + \delta T - \gamma}{\beta + \delta}$

(d) $p^s = \frac{-\beta T + \alpha - \gamma}{\beta + \delta}$

(e) $Q^T = \frac{-\beta\delta T + \delta\alpha + \beta\gamma}{\beta + \delta}$

(f) Government revenue is $\frac{T(\delta(\alpha - \delta T) + \beta\gamma)}{\beta + \delta}$

(g) Consumer incidence is $\frac{\delta}{\beta + \delta}$ and producer incidence is $\frac{\beta}{\beta + \delta}$

(h) $DWL = \frac{\beta\delta T^2}{2(\beta + \delta)}$

1.2 Subsidies

(a) $p^d = \frac{\alpha - \delta S - \gamma}{\beta + \delta}$

(b) $p^s = \frac{\beta S + \alpha - \gamma}{\beta + \delta}$

(c) $Q^S = \frac{\beta\delta S + \delta\alpha + \beta\gamma}{\beta + \delta}$

(d) Government expenditure is $\frac{S(\delta(\delta S + \alpha) + \beta\gamma)}{\beta + \delta}$

(e) Consumer incidence is $\frac{\delta}{\beta + \delta}$ and producer incidence is $\frac{\beta}{\beta + \delta}$, similar to the case of a tax.

(f) $DWL = \frac{\beta\delta S^2}{2(\beta + \delta)}$

1.3 Limit Cases (1)—Taxes and Subsidies

- (a) If $\beta = 0$ then $p^s = p^e$, which implies that the consumer pays the whole tax/receives the whole subsidy, as can be seen by the consumer incidence expression. Furthermore Q^T and Q^S are equal to Q^e .

- (b) If $\delta = 0$ then $p^d = p^e$, which implies that the producer pays the whole amount of the tax/receives the whole subsidy, as can be seen by the producer incidence expression. Furthermore, the dead-weight loss in this case is also zero. Furthermore Q^T and Q^S are equal to Q^e .
- (c) In both the tax and subsidy cases, the dead-weight loss expression becomes zero if either $\beta = 0$ or $\delta = 0$. When demand or supply are perfectly inelastic, the distortion created by the tax or subsidy is zero and therefore the dead-weight loss is zero.
- (d) As mentioned before, when demand is inelastic, the tax incidence/subsidy incidence expression says that the consumer will pay the whole tax/receive the whole subsidy. Similarly, when supply is inelastic, the tax incidence/subsidy incidence expression says that the producer will pay the whole amount of the tax/receive the whole subsidy.

1.4 Limit Cases (2)—Taxes and Subsidies

- (a) Consider first the case of perfectly elastic demand. We have that $Q^e = \gamma + \delta p^*$ and $p^e = p^*$.

	Tax	Subsidy
p^d	p^*	p^*
p^s	$p^* - T$	$p^* + S$
$Q^T Q^S$	$\gamma + \delta p^* - \delta T = Q^e - \delta T$	$\gamma + \delta p^* + \delta S = Q^e + \delta S$
Revenue Expenditure	$T(\gamma + \delta p^* - \delta T)$	$S(\gamma + \delta p^* + \delta S)$
Consumer Incidence Producer Incidence	0 1	0 1
Dead-Weight Loss	$\frac{\delta T^2}{2}$	$\frac{\delta S^2}{2}$

- (b) Now consider the case of perfectly elastic supply. We have that $Q^e = \alpha - \beta p^*$ and $p^e = p^*$.

	Tax	Subsidy
p^d	$p^* + T$	$p^* - S$
p^s	p^*	p^*
$Q^T Q^S$	$\alpha - \beta p^* - \beta T = Q^e - \beta T$	$\alpha - \beta p^* + \beta S = Q^e + \beta S$
Revenue Expenditure	$T(\alpha - \beta p^* - \beta T)$	$S(\alpha - \beta p^* + \beta S)$
Consumer Incidence Producer Incidence	1 0	1 0
Dead-Weight Loss	$\frac{\beta T^2}{2}$	$\frac{\beta S^2}{2}$

1.5 Taxes and Subsidies—Examples and Welfare Analysis

Table 1 provides the calculations for taxes while Table 2 shows the corresponding calculations for subsidies.

From both tables you can check that the identities indeed hold. Consider the following:

- For a tax, where $SS^{\text{before tax}} \geq SS^{\text{after tax}}$, the change in social surplus is $\Delta SS = SS^{\text{before tax}} - SS^{\text{after tax}}$. Thus, using the identity, we can write

$$\Delta SS - DWL = R,$$

which shows that government revenues are given by the change in social surplus minus the dead-weight loss which shows that indeed some part of the surplus is lost irrevocably.

Table 1: Calculations for taxes

	Case I	Case II	Case III	Case IV	Case V
Amount of tax paid by consumers	111.11	1.00	0.00	22,500.00	7.00
Amount of tax paid by producers	222.22	3.00	4,500,000.00	22,500.00	0.00
Government revenue from the tax	333.33	4.00	4,500,000.00	45,000.00	7.00
Consumer surplus before tax	400.00	2.00	81,000,000.00	45,000.00	32.00
Producer surplus before tax	750.00	6.00	9,000,000.00	45,000.00	0.00
Social surplus before tax	1,150.00	8.00	90,000,000.00	90,000.00	32.00
Consumer surplus after tax	277.78	0.50	81,000,000.00	11,250.00	24.50
Producer surplus after tax	505.56	1.50	4,500,000.00	11,250.00	0.00
Social surplus after tax	783.33	2.00	85,500,000.00	22,500.00	24.50
Dead Weight Loss	33.33	2.00	0.00	22,500.00	0.50

Table 2: Calculations for subsidies

	Case I	Case II	Case III	Case IV	Case V
Amount of subsidy received by consumers	155.56	2.06	0.00	100,000.00	180.00
Amount of subsidy received by producers	311.11	6.19	1,800,000.00	100,000.00	0.00
Government expenditure from the subsidy	466.67	8.25	1,800,000.00	200,000.00	180.00
Consumer surplus before subsidy	400.00	2.00	81,000,000.00	45,000.00	32.00
Producer surplus before subsidy	750.00	6.00	9,000,000.00	45,000.00	0.00
Social surplus before subsidy	1,150.00	8.00	90,000,000.00	90,000.00	32.00
Consumer surplus after subsidy	544.44	3.78	81,000,000.00	125,000.00	162.00
Producer surplus after subsidy	1,038.89	11.34	10,800,000.00	125,000.00	0.00
Social surplus after subsidy	1,583.33	15.13	91,800,000.00	250,000.00	162.00
Dead Weight Loss	33.33	1.13	0.00	40,000.00	50.00

- For a subsidy, where $SS^{\text{before subsidy}} \leq SS^{\text{after subsidy}}$, the change in social surplus is $\Delta SS = SS^{\text{after subsidy}} - SS^{\text{before subsidy}}$. Thus, using the identity, we can write

$$\Delta SS + DWL = E,$$

which shows that resources from government expenditures are used to fund the change in social surplus but part of these resources are also being lost in the form of an irrevocable loss.

2 Ad-Valorem Taxation

- (a) $p^d = \frac{(\tau+1)(\alpha-\gamma)}{\beta(\tau+1)+\delta}$
- (b) $p^s = \frac{\alpha-\gamma}{\beta(\tau+1)+\delta}$
- (c) $Q^\tau = \frac{\delta\alpha(\tau+1)+\beta\gamma}{\beta(\tau+1)+\delta}$
- (d) Government revenue is $\frac{\tau(\alpha-\gamma)(\delta\alpha(\tau+1)+\beta\gamma)}{(\beta\tau+\beta+\delta)^2}$
- (e) $DWL = \frac{\beta\delta\tau^2(\alpha-\gamma)^2}{2(\beta+\delta)(\beta\tau+\beta+\delta)^2}$

3 The Agricultural Burden

3.1 The Great Potato Mountain (price supports)

Given:

- Demand: $Q = 100 - 2p$
- Supply: $Q = 10 + p$
- Price support: $\bar{p} = 40$

- (a) Equilibrium price (p^e) and quantity (Q^e) without intervention

Setting demand equal to supply:

$$\begin{aligned} 100 - 2p &= 10 + p \\ 90 &= 3p \\ p^e &= 30 \end{aligned}$$

Substituting back: $Q^e = 10 + 30 = 40$

- (b) Quantity demanded ($\tilde{q}_d$) and supplied ($\tilde{q}_s$) at support price $\bar{p} = 40$

$$\begin{aligned} \tilde{q}_d &= 100 - 2(40) = 20 \\ \tilde{q}_s &= 10 + 40 = 50 \end{aligned}$$

- (c) Government purchases ($\tilde{q}_g$)

$$\tilde{q}_g = \tilde{q}_s - \tilde{q}_d = 50 - 20 = 30$$

(d) **Total government expenditure (X)**

$$X = \bar{p} \times \tilde{q}_g = 40 \times 30 = 1200$$

(e) **Deadweight Loss (DWL)**

We calculate welfare before and after the price support.

Before price support:

Consumer Surplus (triangle):

$$CS_{\text{before}} = \frac{1}{2}(50 - 30) \times 40 = 400$$

Producer Surplus (trapezoid, since supply starts at $Q = 10$ when $p = 0$):

$$PS_{\text{before}} = 10 \times 30 + \frac{1}{2} \times 30 \times (40 - 10) = 300 + 450 = 750$$

Social Surplus:

$$SS_{\text{before}} = 400 + 750 = 1150$$

After price support:

Consumer Surplus (triangle):

$$CS_{\text{after}} = \frac{1}{2}(50 - 40) \times 20 = 100$$

Producer Surplus (trapezoid):

$$PS_{\text{after}} = 10 \times 40 + \frac{1}{2} \times 40 \times (50 - 10) = 400 + 800 = 1200$$

Social Surplus:

$$SS_{\text{after}} = 100 + 1200 = 1300$$

Calculating DWL:

Change in social surplus:

$$\Delta SS = SS_{\text{after}} - SS_{\text{before}} = 1300 - 1150 = 150$$

Using the identity $DWL = X - \Delta SS$:

$$DWL = 1200 - 150 = 1050$$

Verification:

The DWL can be computed geometrically as the sum of three areas representing inefficiency:

$$\begin{aligned} DWL &= \frac{1}{2}(40 - 10)(40 - 20) + \frac{1}{2}(50 - 20)(40 - 10) + (50 - 20) \times 10 \\ &= \frac{1}{2}(30)(20) + \frac{1}{2}(30)(30) + 30 \times 10 \\ &= 300 + 450 + 300 \\ &= 1050 \checkmark \end{aligned}$$

Summary Table:

Measure	Before Support	After Support
Price	$p^e = 30$	$\bar{p} = 40$
Quantity (market)	$Q^e = 40$	$\tilde{q}_d = 20$
Quantity supplied	$Q^e = 40$	$\tilde{q}_s = 50$
Government purchases	0	$\tilde{q}_g = 30$
Consumer Surplus	400	100
Producer Surplus	750	1200
Social Surplus	1150	1300
Government Expenditure	0	$X = 1200$
Change in Social Surplus	$\Delta SS = 150$	
Deadweight Loss	$DWL = 1050$	

3.2 The "Just Kaspi the Money" Approach (Deficiency Payments)

Given the same demand and supply functions with support price $\bar{p} = 40$:

- (a) **Production Mode: Quantity produced ($\hat{q}$) at $\bar{p} = 40$**

Farmers produce based on the support price they're guaranteed:

$$\hat{q} = 10 + \bar{p} = 10 + 40 = 50$$

- (b) **The Fire Sale: Market clearing price (p_2) when $\hat{q} = 50$ is supplied**

The market price must fall to the point where consumers demand the entire quantity produced:

$$100 - 2p_2 = 50$$

$$2p_2 = 50$$

$$p_2 = 25$$

- (c) **The Subsidy: Deficiency payment per unit**

The deficiency payment is the difference between the support price and the market price:

$$\text{Deficiency Payment} = \bar{p} - p_2 = 40 - 25 = 15$$

- (d) **The Comparison: Total Government Expenditure (X)**

The government pays the deficiency payment for each unit produced:

$$X = (\bar{p} - p_2) \times \hat{q} = 15 \times 50 = 750$$

Comparison with Problem 3.1 (Potato Mountain):

- Price support (buying surplus): $X = 1200$
- Deficiency payments: $X = 750$
- Savings: $1200 - 750 = 450$

(e) **Welfare Paradox: DWL and Consumer Surplus**

Before intervention: (same as Problem 3.1)

$$CS_{\text{before}} = 400$$

$$PS_{\text{before}} = 750$$

$$SS_{\text{before}} = 1150$$

After deficiency payments:

Consumer Surplus (consumers pay $p_2 = 25$ for $\hat{q} = 50$ units):

$$CS_{\text{after}} = \frac{1}{2}(50 - 25) \times 50 = 625$$

Producer Surplus (producers receive $\bar{p} = 40$ for all 50 units, trapezoid):

$$PS_{\text{after}} = 10 \times 40 + \frac{1}{2} \times 40 \times (50 - 10) = 400 + 800 = 1200$$

Social Surplus:

$$SS_{\text{after}} = 625 + 1200 = 1825$$

Calculating DWL:

Change in social surplus:

$$\Delta SS = SS_{\text{after}} - SS_{\text{before}} = 1825 - 1150 = 675$$

Using the identity $DWL = X - \Delta SS$:

$$DWL = 750 - 675 = 75$$

Consumer Surplus Change:

$$\Delta CS = CS_{\text{after}} - CS_{\text{before}} = 625 - 400 = 225$$

Consumers are better off by 225 because they pay a lower price ($p_2 = 25$ instead of $p^e = 30$) and consume more ($\hat{q} = 50$ instead of $Q^e = 40$).

Summary Table:

Measure	Before	After Deficiency Payments
Price received by producers	$p^e = 30$	$\bar{p} = 40$
Price paid by consumers	$p^e = 30$	$p_2 = 25$
Quantity produced	$Q^e = 40$	$\hat{q} = 50$
Deficiency payment per unit	0	15
Consumer Surplus	400	625
Producer Surplus	750	1200
Social Surplus	1150	1825
Government Expenditure	0	$X = 750$
Change in Social Surplus		$\Delta SS = 675$
Deadweight Loss		$DWL = 75$

Comparison with Price Supports (Problem 3.1):

Measure	Price Support	Deficiency Payments
Government Expenditure	1200	750
Deadweight Loss	1050	75
Consumer Surplus Change	-300	+225
Efficiency Gain	Deficiency payments save \$450 and reduce DWL by \$975	

4 Trade Policy: Protecting the Inefficient

4.1 The Chicken Toll Booth (Import Tariffs)

Given:

- Domestic demand: $Q_D = 100 - 2p$
- Domestic supply: $Q_S = 10 + p$
- World price: $p^* = 20$
- Tariff: $T = 5$

(a) Free Trade: Domestic production (Q_1), consumption (Q^F), and imports

At the world price $p^* = 20$:

Domestic production:

$$Q_1 = 10 + p^* = 10 + 20 = 30$$

Domestic consumption:

$$Q^F = 100 - 2p^* = 100 - 2(20) = 60$$

Imports:

$$\text{Imports} = Q^F - Q_1 = 60 - 30 = 30$$

(b) The Protection: New domestic price with tariff $T = 5$

With the tariff, the domestic price rises to:

$$p_{\text{domestic}} = p^* + T = 20 + 5 = 25$$

(c) The Resulting Distortion: New domestic production (Q_2) and imports

At the new domestic price $p = 25$:

Domestic production:

$$Q_2 = 10 + 25 = 35$$

Domestic consumption:

$$Q_3 = 100 - 2(25) = 50$$

New imports:

$$\text{Imports} = Q_3 - Q_2 = 50 - 35 = 15$$

Imports fall from 30 units to 15 units (a reduction of 15 units or 50%).

(d) The Payday: Government Revenue

Government revenue comes from the tariff on imports:

$$R = T \times \text{Imports} = 5 \times 15 = 75$$

(e) The Accounting: Deadweight Loss triangles

Area C (Production Distortion): Triangle from $Q_1 = 30$ to $Q_2 = 35$

Base: $Q_2 - Q_1 = 5$, Height: $T = 5$

$$C = \frac{1}{2} \times 5 \times 5 = 12.5$$

Area E (Consumption Loss): Triangle from $Q_3 = 50$ to $Q^F = 60$

Base: $Q^F - Q_3 = 10$, Height: $T = 5$

$$E = \frac{1}{2} \times 10 \times 5 = 25$$

Total Deadweight Loss:

$$DWL = C + E = 12.5 + 25 = 37.5$$

Welfare Analysis:

Before Tariff (Free Trade):

Consumer Surplus:

$$CS_{\text{before}} = \frac{1}{2} (50 - 20) \times 60 = 900$$

Producer Surplus (trapezoid, supply starts at $Q = 10$ when $p = 0$):

$$PS_{\text{before}} = 10 \times 20 + \frac{1}{2} \times 20 \times (30 - 10) = 200 + 200 = 400$$

Social Surplus:

$$SS_{\text{before}} = 900 + 400 = 1300$$

After Tariff:

Consumer Surplus:

$$CS_{\text{after}} = \frac{1}{2}(50 - 25) \times 50 = 625$$

Producer Surplus (trapezoid):

$$PS_{\text{after}} = 10 \times 25 + \frac{1}{2} \times 25 \times (35 - 10) = 250 + 312.5 = 562.5$$

Social Surplus:

$$SS_{\text{after}} = 625 + 562.5 = 1187.5$$

Verification:

Change in Social Surplus:

$$\Delta SS = 1187.5 - 1300 = -112.5$$

Identity check:

$$\Delta SS + R = -112.5 + 75 = -37.5 = -DWL \checkmark$$

Summary Table:

Measure	Free Trade	With Tariff
Domestic Price	$p^* = 20$	25
Domestic Production	$Q_1 = 30$	$Q_2 = 35$
Domestic Consumption	$Q^F = 60$	$Q_3 = 50$
Imports	30	15
Consumer Surplus	900	625
Producer Surplus	400	562.5
Social Surplus	1300	1187.5
Government Revenue	0	$R = 75$
Change in SS	$\Delta SS = -112.5$	
Production Distortion (C)	12.5	
Consumption Loss (E)	25	
Deadweight Loss	$DWL = 37.5$	

4.2 The Quota Identity Crisis

(a) **Equivalence: Equivalent quota level ($\bar{Q}$) that results in price of 25**

From problem 4.1, we found that with a tariff, the domestic price rises to 25, resulting in imports of 15 units.

For a quota to achieve the same domestic price of 25, it must restrict imports to:

$$\bar{Q} = Q_3 - Q_2 = 15 \text{ units}$$

This quota will cause the domestic price to rise to 25 (same as the tariff) because at $p = 25$:

- Domestic demand: $Q_3 = 50$ units
- Domestic supply: $Q_2 = 35$ units
- Gap (imports needed): $50 - 35 = 15$ units = $\bar{Q}$

(b) Foreign Aid: Quota rent if licenses given to foreign firms

When the quota is binding, there's a difference between the world price and the domestic price. This creates a "quota rent":

Quota rent per unit:

$$\text{Quota rent per unit} = p_{\text{domestic}} - p^* = 25 - 20 = 5$$

Total quota rent:

$$\text{Total Quota Rent (Area D)} = 5 \times 15 = 75$$

If the government gives the import licenses to foreign firms for free, these firms can buy at the world price (20) and sell in the domestic market at the domestic price (25), capturing the rent of 75.

This 75 leaves the country as profit to foreign firms.

(c) National Welfare: Comparison of Social Surplus

Under Tariff (from 4.1):

- $CS = 625$
- $PS = 562.5$
- $SS = 1187.5$
- Government revenue: $R = 75$ (stays in country)
- DWL (triangles C + E): 37.5
- Net domestic welfare change: $1187.5 + 75 - 1300 = -37.5$

Under Quota with Foreign Licenses:

- $CS = 625$ (same, because domestic price is same)
- $PS = 562.5$ (same, because domestic production and price are same)
- $SS = 1187.5$ (same)
- Government revenue: 0 (no tariff collected)
- Quota rent to foreigners: 75 (leaves the country)
- DWL (triangles C + E + area D): $37.5 + 75 = 112.5$
- Net domestic welfare change: $1187.5 - 1300 = -112.5$

Comparison:

Change in domestic welfare from tariff to quota:

$$\Delta W = -112.5 - (-37.5) = -75$$

The country is worse off by 75 under the quota (with foreign licenses) compared to the tariff.

Measure	Tariff	Quota (Foreign Licenses)
Domestic Price	25	25
Imports	15	$\bar{Q} = 15$
Consumer Surplus	625	625
Producer Surplus	562.5	562.5
Social Surplus	1187.5	1187.5
Government Revenue	$R = 75$	0
Quota Rent (to foreigners)	0	75
DWL (C + E)	37.5	37.5
Additional DWL (Area D)	0	75
Total DWL	37.5	112.5
Net Domestic Welfare Change	-37.5	-112.5

Summary Table:

Key Insight:

Under both policies, consumers and domestic producers experience identical effects (same prices and quantities). The critical difference is:

- **Tariff:** Government collects 75 in revenue (stays in the country) $\rightarrow$ DWL = 37.5
- **Quota with foreign licenses:** Foreign firms capture 75 in quota rents (leaves the country) $\rightarrow$ DWL = 112.5

Therefore, the quota results in an additional welfare loss of 75 to the domestic economy compared to the equivalent tariff. The quota rent (Area D) becomes part of the deadweight loss from the domestic country's perspective.